ALEK FRÖHLICH

Ph.D. Student in Machine Learning

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EDUCATION

ELLIS Ph.D. in Computer Science — Nov 2024 – Present
Istituto Italiano di Tecnologia & École Polytechnique — Genoa, Italy

- Leveraging low-rank spectral structure for scalable and statistically rigorous causal inference with high-dimensional data.
- Advisors: Massimiliano Pontil, Karim Lounici, Vladimir Kostić.

M.Sc. in Applied Mathematics — Mar 2022 – Jul 2024
Federal University of Santa Catarina — Florianópolis, Brazil

- GPA: 9.76/10. Courses: Functional analysis, machine learning theory, numerical optimization, differential equations, topology, advanced analysis, linear algebra and matrix theory.
- Dissertation: Elements of Learning theory and their Application in the Prediction of Malignancy of Breast Lesions.

B.Sc. in Computer Science — Mar 2018 – Mar 2022
Federal University of Santa Catarina — Florianópolis, Brazil

- GPA: 9.23/10. (2nd in cohort). Some courses: Measure theory, machine learning, probability, statistics, parallel programming, algorithms, software engineering, operating systems.

Exchange Student — Aug 2016 - Dec 2016
University High School — Irvine, CA

- AGPA: 4.2. AP Course: Computer science.

EXPERIENCE

Research Intern — Jul 2024 – Sep 2024
Federal University of São Carlos — São Carlos, Brazil

- Developed an interpretable breast cancer risk assessment tool with local coverage uncertainty quantification guarantees via conformal prediction. Resulted in [4]. Mentor: Rafael Izbicki.

Research Intern — Jul 2023 – Jul 2024
University of São Paulo Medical School — Ribeirão Preto, Brazil

- Developed and validated machine learning models for predicting malignancy of suspicious breast lesions identified during ultrasound examinations. Resulted in [5].
- Developed a computational pathology pipeline for segmenting and classifying tumor cells in whole slide images. Resulted in [6]. Mentor: Daniel Tiezzi.

Visiting Student — Jan 2023 – Nov 2023
Institute for Pure and Applied Mathematics (IMPA) — Rio de Janeiro, Brazil

- Courses: Functional analysis and machine learning theory.
- Seminars: Optimization (mentored by Benar Svaiter) and physics-informed neural networks.

Software Engineer Intern Jan 2019 – Dec 2019
LabSEC & Kryptus Florianópolis, Brazil

PUBLICATIONS & PRE-PRINTS

Symbols: † denotes equal contribution, * denotes highlight.

1. T Ramos, **A Fröhlich**, D Perazzo, M Pontil. *Drift-Aware Uncertainty Quantification via a Functional Spectral-Newton Method*. ICLR Workshop CAO 2026. [OpenReview]
2. * **A Fröhlich**, V Kostić, K Lounici, D Perazzo, M Pontil. *Toward Scalable and Valid Conditional Independence Testing with Spectral Representations*. 2025. [arXiv]
3. * D Meunier,† J Wornbard,† V Kostić,† A Moulin, **A Fröhlich**, K Lounici, M Pontil, A Gretton. *Outcome-Aware Spectral Feature Learning for Instrumental Variable Regression*. 2025. [arXiv]
4. D Ordoñez-Apraez, V Kostić, **A Fröhlich**, V Brandt, K Lounici, M Pontil. *Equivariant Representation Learning for Symmetry-Aware Inference with Guarantees*. 2025. [arXiv]
5. **A Fröhlich**,† T Ramos,† G Santos, I Buzatto, R Izbicki, D Tiezzi. *PersonalizedUS: Interpretable Breast Cancer Risk Assessment with Local Coverage Uncertainty Quantification*. AAAI 2025. [DOI]
6. I Buzatto, S Recife, L Miguel, R Bonini, N Onari, A Faim, L Silvestre, D Carlotti, **A Fröhlich**, D Tiezzi. *Machine Learning can Reliably Predict Malignancy of Breast Lesions based on Clinical and Ultrasonographic Features*. Breast Cancer Research and Treatment 2024. [DOI]
7. D Tiezzi, **A Fröhlich**, F Chahud, S Pagnota. *197P Computational pathology pipeline enables quantification of intratumor heterogeneity and tumor-infiltrating lymphocyte score*. ESMO Immuno-Oncology and Technology 2023. [DOI]

TALKS

Spectral Representation Learning Oct 2025
University of Genoa Genoa, Italy

Spectral Representation Learning for High-dimensional Inference Aug 2025
Federal University of São Carlos São Carlos, Brazil

Breast Cancer Risk Assessment with Local Coverage UQ Jun 2025
New Trends in Statistical Learning V Porquerolles, France

Machine Learning Methods in Her2+ Breast Cancer Oct 2023
Institute for Pure and Applied Mathematics (IMPA) Rio de Janeiro, Brazil

AI and Her2+ Breast Cancer Aug 2023
University of São Paulo Medical School Ribeirão Preto, Brazil

SVM: Optimization Problem, Learning Guarantees, and Kernel Trick Jul 2023
34th Brazilian Mathematical Colloquium Rio de Janeiro, Brazil

SUPERVISIONS

Enzo N. Spotorno (B.Sc. co-advised with A. A. Fröhlich) Jul 2024 – Jul 2025
Federal University of Santa Catarina
Project: Improving Battery End-of-Life Prediction with Hybrid Bayesian PINNs

Vagner S. Santos (M.Sc. co-advised with R. Izbicki) Mar 2026 – Mar 2028
Federal University of São Carlos
Project: Spectral Boosting for Nonparametric Causal Inference

AWARDS & RECOGNITIONS

G-Research Early Career Grant 2026
G-Research UK

- Competitive grant awarded to support a research exchange at École Polytechnique (£2,000).

EMS Membership 2026 – 2029
European Mathematical Society (*EMS*) Europe

- Student member of EMS until February 2029.

INdAM Membership (Real Analysis, Measure Theory & Probability) 2026
Istituto Nazionale di Alta Matematica (*INdAM*) Italy

- National network of mathematical excellence supporting research projects and mobility.

CAPES Master's Scholarship 2022 – 2024
Brazilian Federal Agency for Support and Evaluation of Graduate Education (*CAPES*) Brazil

- Fully funded my M.Sc. studies in Applied Mathematics (R$ 50,400).

FAPEU Scholarship 2019
Fundação de Amparo à Pesquisa e Extensão Universitária (*FAPEU*) Brazil

- Awarded for developing the access control system for UFSC's university restaurant.

FEESC Scholarship 2019
Fundação Stemmer para Pesquisa, Desenvolvimento e Inovação (*FEESC*) Brazil

- Awarded for work on the Cryptographic Service Provider developed at LabSEC.

ITI Scholarship 2019
Instituto Nacional de Tecnologia da Informação (*ITI*) Brazil

- Awarded for work on the Cryptographic Service Provider developed at LabSEC.

OUTREACH & LEADERSHIP

Reviewer for ICLR 2026 Workshop: Catch, Adapt, and Operate Feb 2026

EVENTS

Computational Imaging & Learning Winter School Feb 2026
University of Genoa Genoa, Italy

Causality for Impact Workshop EurIPS 2025 Dec 2025
Oral Presentation [2] Copenhagen, Denmark

Mathematical Aspects of Data Science Summer School Sep 2025
École Polytechnique Fédérale de Lausanne (*EPFL*) Lausanne, Switzerland

Workshop on Making Sense of Data in Robotics CoRL 2025 — Sep 2025
Presentation of [4] — Seoul, Korea

Theoretical Foundations of Machine Learning Summer School — Jun 2025
University of Genoa — Genoa, Italy

New Trends in Statistical Learning V — Jun 2025
Presented [5] — Porquerolles, France

39th Annual AAAI Conference on Artificial Intelligence — Feb 2025
Presented [5] — Philadelphia, PA

Workshop on Optimization and Inverse Problems — 2023
Federal University of Santa Catarina — Florianópolis, Brazil

34th Brazilian Mathematical Colloquium — 2023
Institute for Pure and Applied Mathematics (IMPA) — Rio de Janeiro, Brazil

Machine Learning and Combinatorics Workshop — 2020
Moscow Institute of Physics and Technology — Moscow (online), Russia

Brazilian Symposium on Computing Systems Engineering IX — 2019
Federal University of Rio Grande do Norte — Natal, Brazil

The Developer's Conference — 2019
Largest event related to software development in Brazil — Florianópolis, Brazil

Chip in the Pampa 2018 — 2018
Federal University of Pelotas — Pelotas, Brazil

Workshop on Quantum Computing I — 2018
Federal University of Santa Catarina — Florianópolis, Brazil

SKILLS

Languages	Portuguese (native), English (fluent), Italian (basic)
Programming	C/C++, Java, Python, R, Matlab, PyTorch
Miscellaneous	Git, Docker, LaTeX, SQL, Assembly (x86, ARM, RISC-V, MIPS)